5 Planning

Iodide ions, I^-, react with peroxodisulfate ions, $S_2O_8^{2-}$, as shown in equation 5.

equation 5 $S_2O_8^{2-}(aq) + 2I^-(aq) \rightarrow 2SO_4^{2-}(aq) + I_2(aq)$

If $I^-(aq)$ is used in large excess in the reaction, the simplified rate equation becomes

$$\text{rate} = k'[S_2O_8^{2-}]^a \qquad \text{where } k' = k[I^-]$$

The order of reaction with respect to peroxodisulfate ions can be determined experimentally using the continuous method. Since iodine (I_2) is formed during the reaction, the concentration of iodine can be measured continuously at various time intervals by sampling, quenching using cold water, followed by titration with aqueous sodium thiosulfate.

(a) You will prepare a reaction mixture consisting of

- 70 cm^3 of 0.400 mol dm^{-3} aqueous potassium iodide, KI
- 30 cm^3 of 0.010 mol dm^{-3} aqueous peroxodisulfate ions, $S_2O_8^{2-}$

Plan an investigation to determine the order of reaction with respect to peroxodisulfate, $S_2O_8^{2-}$, using the continuous method.

In your plan you should include brief details of:

- the quantities of all reagents you would use,
- the apparatus and the procedure you would follow,
- the measurements you would take to allow for a suitable volume-time graph to be drawn from time, t = 0 min to t = 20 min.

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(b) (i) V_{max} represents the volume of sodium thiosulfate required to react with the iodine produced when the reaction as shown by equation 5 has completed, in a 10.0 cm^3 aliquot of the reaction mixture.

$$I_2 + 2S_2O_3^{2-} \rightarrow 2I^- + S_4O_6^{2-}$$

Show that V_{max} is 20.00 cm^3.

[2]

(ii) Using information from **(b) (i)**, sketch on Fig. 5.1 an appropriately labelled graph to show that the reaction is first order with respect to peroxodisulfate ions.

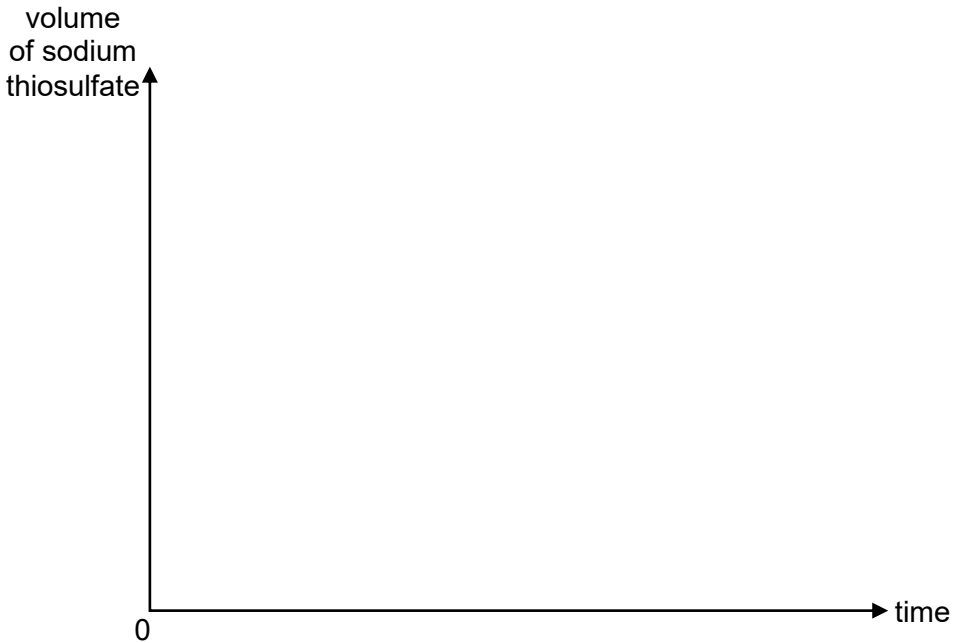

Fig. 5.1

[2]

(c) The reaction between peroxodisulfate ions and iodide ions as shown in equation 5 is slow. The reaction can be sped up by the addition of iron(II) catalyst, Fe^{2+} catalyst.

(i) Suggest why the reaction between peroxodisulfate ions and iodide ions is slow at room temperature.

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(ii) The experimental procedure in **(a)** is modified to determine the order of reaction with respect to the iron(II) catalyst in the reaction between peroxodisulfate and iodide ions.

Two additional experiments were conducted with iron(II) catalyst added at the beginning.

You are provided with 50 cm^3 of 0.01 mol dm^{-3} iron(II) nitrate solution, $Fe(NO_3)_2$.

- Complete Table 5.1 with suitable volumes of reagents used.
- Describe how you would make use of the data collected to determine the order of reaction with respect to the iron(II) catalyst.

Table 5.1

expt	volume of $S_2O_8^{2-}$ / cm^3	volume of I^- / cm^3	volume of H_2O / cm^3	volume of Fe^{2+} / cm^3
1	30	70		
2	30	70		

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